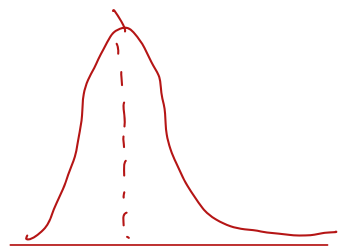


Consistencia

¿Qué tan lejos es probable que esté el estimador, $\hat{\theta}_n$, del parámetro verdadero, θ , a medida que el tamaño de la muestra aumenta?



$$\Pr(|W_n - \theta| > \epsilon) \rightarrow 0 \text{ cuando } n \rightarrow \infty$$

$$\lim_{n \rightarrow \infty} \Pr(|W_n - \theta| > \epsilon) = 0$$

$$\text{plim}(W_n - \theta) = 0 \quad \begin{array}{l} \text{plim}(W_n) = \theta \\ \text{plim}(\hat{\theta}) = \theta \end{array}$$

Ley DE los grandes números

X es una variable aleatoria

$E(X) \rightarrow$ es el primer momento

$\bar{X}_n \rightarrow$ es el promedio muestral.

$$\bar{X}_n = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$$

A medida que N aumenta $\bar{X}_n$ se aproxima a $E(X)$

Sea $X_1, X_2, \dots, X_n$ diferentes realizaciones de una variable

aleatoria X_n que se distribuye iid con $E(X) = \mu$ tenemos que

$$\text{plim} \left(\frac{1}{n} \sum X_n \right) = E(X) = \mu$$

$$\text{plim} \left(\frac{1}{n} \sum X_i Y_i \right) = E[X_i Y_i]$$

Propiedades asintóticas de los MCO

$$\hat{\beta} = (X'X)^{-1} X'y$$

$$= (X'X)^{-1} X'[X\beta + e]$$

$$\frac{(\hat{\epsilon}'_1 \hat{\epsilon}_1 - \hat{\epsilon}' \hat{\epsilon}) / \#r}{\frac{\hat{\epsilon}' \hat{\epsilon}}{n-K}}$$

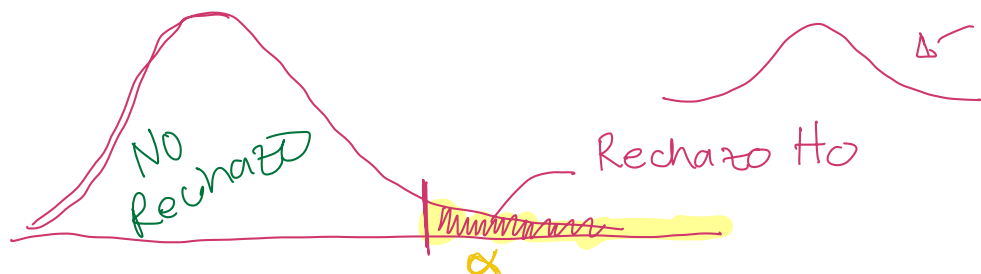
$$y = \beta_0 + \epsilon_r$$

$$\hat{\beta}_0 = \bar{y} \quad \hat{y} = \bar{y} \quad \hat{\epsilon}_r = y - \bar{y}$$

$$\begin{aligned} \hat{\epsilon}'_1 \hat{\epsilon}_1 &= (y - \bar{y})'(y - \bar{y}) \\ &= \sum (y_i - \bar{y})^2 \\ &= STC_{NR} \end{aligned}$$

$$\frac{\frac{(STC_{NR} - SRC_{NR}) / STC}{K-1}}{\frac{SRC_{NR} / STC}{n-K}}$$

$$\frac{\left(\frac{1 - \frac{SRC_{NR}}{STC}}{K-1} \right)}{\frac{(1 - R^2)}{n-K}} = \frac{R^2 / K-1}{(1 - R^2) / n-K}$$



$$= \beta + (X'X)^{-1} X' \varepsilon \frac{n}{n}$$

$$\hat{\beta} - \beta = \left(\frac{X'X}{n} \right)^{-1} \frac{X'\varepsilon}{n}$$

$$\text{plim} (\hat{\beta} - \beta) = \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim} \left(\frac{X'\varepsilon}{n} \right)$$

$$\text{plim} (\hat{\beta}) = \text{plim} (\beta) + \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim} \left(\frac{X'\varepsilon}{n} \right)$$

$\rightarrow \beta$
 $\rightarrow E[X'X]^{-1} E[X'\varepsilon]$
 $\rightarrow Q^{-1} \rightarrow 0$
 $\rightarrow 0$

$$\hat{\sigma}_{\varepsilon}^2 = \frac{\hat{\varepsilon}' \hat{\varepsilon}}{n-K} \frac{n}{n}$$

$$\text{plim} (\hat{\sigma}_{\varepsilon}^2) = \text{plim} \left(\frac{n}{n-K} \right) \text{plim} \left(\frac{\hat{\varepsilon}' \hat{\varepsilon}}{n} \right)$$

$\rightarrow 1$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$y = 0.5 + 0.3x_1 + \varepsilon$$

$$= \beta + (X'X)^{-1}X'\varepsilon$$

$$\rightarrow E[\hat{\beta}|X] = E[\beta + \underbrace{(X'X)^{-1}X'\varepsilon}_{=0}] = \beta \quad \text{por supuesto 2}$$

$$\hat{\beta} = (X'X)^{-1}X'y \frac{n}{n}$$

$$= \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'y}{n} \right)$$

$$= \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'(X\beta + \varepsilon)}{n} \right)$$

$$= \left(\frac{X'X}{n} \right)^{-1} \left[\left(\frac{X'X}{n} \right) \beta + \frac{X'\varepsilon}{n} \right]$$

$$= \cancel{\left(\frac{X'X}{n} \right)^{-1}} \cancel{\left(\frac{X'X}{n} \right)} \beta + \left(\frac{X'X}{n} \right)^{-1} \frac{X'\varepsilon}{n}$$

$$= \beta + \left(\frac{X'X}{n} \right)^{-1} \frac{X'\varepsilon}{n}$$

$$\hat{\beta} = (X'X)^{-1} X'y = \beta + (X'X)^{-1} X'e$$

$$\hat{\beta} = \beta \frac{n}{n} + (X'X)^{-1} X'e \frac{n}{n}$$

$$\hat{\beta} = \beta + \left(\frac{X'X}{n} \right)^{-1} \frac{X'e}{n}$$

$$\begin{aligned} \text{plim} [\hat{\beta}] &= \text{plim} [\beta] + \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim} \left(\frac{X'e}{n} \right) \\ &= \beta + \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim} \left(\frac{X'e}{n} \right) \end{aligned}$$

$$\begin{array}{ccc} \xrightarrow{\quad} & \xrightarrow{\quad} & \\ E[X'X/n]^{-1} & E[X'e/X] & \\ \neq 0 & = 0 & \text{se } \text{Cov}(X, e) = 0 \\ \mathbb{Q}^{-1} & & \end{array}$$

$$\text{plim}[\hat{\beta}] = \beta \Rightarrow \text{Consistencia}$$

$$\text{plim} [\hat{\sigma}_e^2] = \sigma_e^2$$

$$\hat{\sigma}_e^2 = \frac{\hat{e}'\hat{e}}{n-K} \cdot \frac{n}{n}$$

$$= \text{plim} \left(\frac{n}{n-k} \right) \text{plim} \left(\frac{\hat{e}'\hat{e}}{n} \right)$$

$$\hat{e} = y - X\hat{\beta} \quad \hat{e}'\hat{e} = (y - X\hat{\beta})'(y - X\hat{\beta})$$

$$\hat{e}'\hat{e} = (X\beta + e - X\hat{\beta})'(X\beta + e - X\hat{\beta})$$

$$\hat{e}'\hat{e} = (e + X(\beta - \hat{\beta}))'(e + X(\beta - \hat{\beta}))$$

$$\text{plim} \left(\frac{\hat{e}'\hat{e}}{n} \right) = \text{plim} \left(\frac{(e + X(\beta - \hat{\beta}))'(e + X(\beta - \hat{\beta}))}{n} \right)$$

$$\text{plim} \left(\frac{e'e}{n} \right) + \text{plim} \left(\frac{e'X}{n} \right) \text{plim}(\beta - \hat{\beta}) + \text{plim}(\beta - \hat{\beta})' \text{plim} \left(\frac{X'e}{n} \right)$$

$\begin{matrix} = 0 & = 0 & = 0 & = 0 \\ & = 0 & & = 0 \end{matrix}$

$$+ \text{plim}(\beta - \hat{\beta})' \text{plim} \left(\frac{X'X}{n} \right) \text{plim}(\beta - \hat{\beta})$$

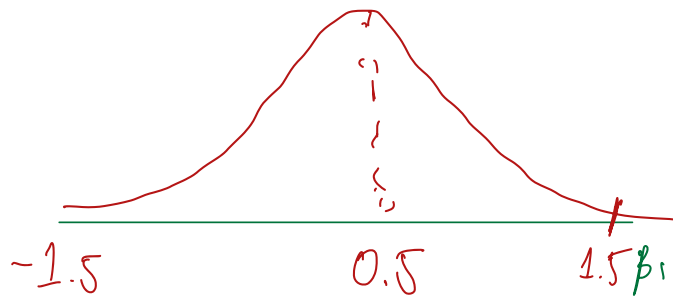
$\begin{matrix} = 0 & \neq 0 & = 0 \\ & = Q & \end{matrix}$

$$\text{plim} \left(\frac{e'e}{n} \right) = E[e'e] = \sigma_e^2$$

Consistent $\hat{\theta}$

$$y = \overset{0.5}{\beta_0} + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$E[\hat{\beta}_1] = \beta_1 \quad \text{Insensado}$$



Rep ~~100~~ 1000

$$\hat{\beta} = (x'x)^{-1}x'y$$

$$E[\hat{\beta}] = \beta \quad \text{Insensado}$$

$$plim[\hat{\beta}] = \beta \quad \text{Consistente}$$

$$\hat{\sigma}_\varepsilon^2 = \frac{\hat{\varepsilon}'\hat{\varepsilon}}{n-K}$$

$$\text{Sesgado} \quad E[\hat{\sigma}_\varepsilon^2] \neq \sigma_\varepsilon^2$$

$$plim[\hat{\sigma}_\varepsilon^2] = \sigma_\varepsilon^2 \quad \text{Consistente.}$$

$$y = X\beta + \epsilon$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$y = X_1\beta_1 + X_5\beta_5 + \epsilon$$

$$\beta_1 \quad y \quad \beta_5$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$y = X_1\beta_1 + X_5\beta_5 + \epsilon$$

$$\textcircled{\beta_1} \rightarrow$$

$$M_5 y = M_5 X_1 \beta_1 + \cancel{M_5 X_5 \beta_5} + \epsilon$$

$$\hat{\beta}_1 = ?$$

$$y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

$$M_2 y = M_2 X_1 \beta_1 + \epsilon$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & \dots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2n} \end{bmatrix} \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \\ x_{13} & x_{23} \\ \vdots & \vdots \\ x_{1n} & x_{2n} \end{bmatrix}$$

$$y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$y = \beta_1 x_1 + \mu$$

$$\hat{\mu}' \hat{\mu} \geq \hat{\varepsilon}' \hat{\varepsilon}$$

$$\mu = [\beta_2 x_2 + \varepsilon]$$

$$\hat{\mu} = \hat{\beta}_2 x_2 + \hat{\varepsilon}$$

$$\text{Rango}(X) < K$$

$$y = X\beta + e$$

$$\hat{\beta} = (X'X)^{-1} X'y$$

$$E[\hat{\beta} | X] = E[\hat{\beta}] = \beta$$

insesgado $\cup$

$$\text{Var}[\hat{\beta} | X] = \sigma_e^2 (X'X)^{-1}$$

$$S^2 = \hat{\sigma}_e^2 = \frac{\text{SRC}}{n-K} = \frac{\hat{\epsilon}'\hat{\epsilon}}{n-K} (X'X)^{-1}$$

$$\underbrace{\frac{\hat{\epsilon}'\hat{\epsilon}}{n-K}}_{1 \times 1} \underbrace{(X'X)^{-1}}_{K \times K}$$

class 1

$$y = X\beta + \epsilon$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} 1 & x_{11} & x_{21} & \dots & x_{k1} \\ 1 & x_{12} & x_{22} & \dots & x_{k2} \\ 1 & x_{13} & x_{23} & \dots & x_{k3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \dots & x_{kn} \end{bmatrix}_{n \times k} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}_{k \times 1}$$

$$y = \beta_0 + \beta_1 x_1 + \epsilon$$

$$\hat{\beta} = \begin{bmatrix} \bar{y} - \hat{\beta}_1 \bar{x} \\ \text{Cov}(x_1, y) / \text{Var}(x_1) \end{bmatrix}$$

$$E[\hat{\beta}] = \beta$$

$$\text{Var}[\hat{\beta}] = \sigma_\epsilon^2 (X'X)^{-1}$$

$$= \begin{bmatrix} \text{Var}(\beta_0) & \text{Cov}(\beta_0, \beta_1) \\ \text{Cov}(\beta_0, \beta_1) & \text{Var}(\beta_1) \end{bmatrix}$$

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1} X'y \\ &= (X'X)^{-1} X' [X\beta + \varepsilon] \\ &= (X'X)^{-1} X'X\beta + \underbrace{(X'X)^{-1} X'\varepsilon} \\ &= \beta + (X'X)^{-1} X'\varepsilon \end{aligned}$$

$$\text{Var}(\hat{\beta} | X) = \text{Var} \left[\beta + \underbrace{(X'X)^{-1} X'\varepsilon}_A \right]$$

$$= A \text{Var}(\varepsilon) A'$$

$$= (X'X)^{-1} X' \underbrace{\sigma_\varepsilon^2 I_n}_{\sigma_\varepsilon^2 I_n} X (X'X)^{-1}$$

$$= \sigma_\varepsilon^2 (X'X)^{-1}$$

Consistencia de Beta

$$E[\hat{\beta}] = \beta$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\text{plim}[\hat{\beta}] = \beta$$

$$\hat{\beta} = \left(\beta + (X'X)^{-1}X'\epsilon \right) \frac{n}{n}$$

$$\text{plim}(\hat{\beta}) = \text{plim}\left(\beta \frac{n}{n}\right) + \text{plim}\left(\left(\frac{X'X}{n}\right)^{-1}\right) \text{plim}\left(\frac{X'\epsilon}{n}\right)$$

$\xrightarrow{\beta} \beta \quad \xrightarrow{E[X'X]^{-1}} E[X'X]^{-1} \quad \xrightarrow{E[X'\epsilon]} E[X'\epsilon] = 0$

Consistencia de σ_{ϵ}^2

$$\hat{\sigma}_{\epsilon}^2 = \frac{\hat{\epsilon}'\hat{\epsilon}}{N-K} = \frac{\hat{\epsilon}'\hat{\epsilon}}{N-K} \cdot \frac{N}{N} = \frac{N}{N-K} \cdot \left(\frac{\hat{\epsilon}'\hat{\epsilon}}{N} \right)$$

$\xrightarrow{\frac{N}{N-K} \rightarrow 1} \xrightarrow{\frac{\hat{\epsilon}'\hat{\epsilon}}{N} \rightarrow \sigma_{\epsilon}^2} \sigma_{\epsilon}^2$

$$= \frac{(y - \hat{y})'(y - \hat{y})}{N}$$

$$= \frac{(X\beta + \epsilon - X\hat{\beta})'(X\beta + \epsilon - X\hat{\beta})}{N}$$

$$= \frac{(\epsilon + X(\beta - \hat{\beta}))'(\epsilon + X(\beta - \hat{\beta}))}{N}$$

$$= \frac{\epsilon'\epsilon}{n} + \frac{\epsilon'X(\beta - \hat{\beta})}{n} + \frac{(\beta - \hat{\beta})'X'\epsilon}{n} + \frac{(\beta - \hat{\beta})'X'X(\beta - \hat{\beta})}{n}$$

$\xrightarrow{E[\epsilon'\epsilon]} E[\epsilon'\epsilon] \quad \xrightarrow{E[\epsilon'X]} 0 \quad \xrightarrow{0} 0 \quad \xrightarrow{E[X'X]} 0$

$$- \quad S^2 = \hat{\sigma}_e^2 = \frac{\hat{\varepsilon}' \hat{\varepsilon}}{n-K} \quad \checkmark \quad = \frac{\sum \varepsilon_i^2}{n-K} \quad \checkmark$$

$$\text{plim}[\hat{\sigma}_e^2] = \sigma_e^2 \quad \text{consistente}$$

$$E[\hat{\sigma}_e^2] = \sigma_e^2$$

$$= E[\epsilon' \epsilon] \\ = \text{Var}(\epsilon) = \sigma_\epsilon^2 \quad \checkmark \checkmark$$

Q

Normalidad asintótica

Slutsky

$$\hat{\beta} \sim N(\beta, \sigma_\epsilon^2 (X'X)^{-1})$$

$$Z_n \xrightarrow{d} Z \text{ y } A_n \xrightarrow{p} \alpha \quad Z_n + A_n \xrightarrow{d} Z + \alpha$$

$$Z_n \xrightarrow{d} Z \text{ y } A_n \xrightarrow{p} 0 \quad Z_n' A_n \xrightarrow{p} 0$$

$$Z_n \xrightarrow{d} Z \text{ y } A_n \xrightarrow{p} A \quad Z_n' A_n \xrightarrow{d} Z' A$$

Ejemplo: Si $Z_n \xrightarrow{d} N(0, \Sigma)$ y $A_n \xrightarrow{p} A$ $Z_n' A_n \xrightarrow{d} N(0, A \Sigma A')$

Teorema central del limite.

Sea $x_1, x_2, \dots, x_n$ realizaciones de una variable aleatoria iid con $E[x_i] = \mu$ $\text{Var}(x_i) = \sigma_x^2 < \infty$ entonces

²Recuerde que el teorema central del limite de Lindeberg-Levy dice:

Sea $X_1, \dots, X_n$ una muestra independiente e idénticamente distribuida con $E[X_i] = \mu$ y $\text{Var}(X_i) = \sigma^2 < \infty$. Entonces:

$$\sqrt{n} \left(\frac{\bar{X} - \mu}{\sigma} \right) \xrightarrow{d} N(0, 1) \text{ cuando } n \rightarrow \infty$$

$$\sqrt{n}(\bar{x} - \mu) \xrightarrow{d} N(0, \sigma_x^2) \quad \frac{\bar{x} - \mu}{\sqrt{n}} \rightarrow N(0, \sigma^2)$$

56. ~~$\epsilon \sim N(0, \sigma_\epsilon^2)$~~
 $\epsilon \sim \text{iid}(0, \sigma_\epsilon^2)$
 $\hat{\beta} \sim N(\beta, \sigma_\epsilon^2 (X'X)^{-1})$

$$\begin{aligned}
 \hat{\beta} &= (X'X)^{-1}X'y \\
 &= (X'X)^{-1}X'[X\beta + \varepsilon] \\
 &= \beta + (X'X)^{-1}X'\varepsilon
 \end{aligned}$$

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon \quad \frac{n}{\sqrt{n}\sqrt{n}}$$

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{X'X}{n} \right)^{-1} \frac{X'\varepsilon}{\sqrt{n}}$$

$$\begin{aligned}
 \text{plim}(\sqrt{n}(\hat{\beta} - \beta)) &= \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim} \left(\frac{X'\varepsilon}{\sqrt{n}} \right) \\
 &\xrightarrow{P} E[X'X]^{-1} \xrightarrow{D} N(0, \underbrace{\sigma_\varepsilon^2 E[X'X]}_{\Sigma})
 \end{aligned}$$

$$A = E[X'X]^{-1} \quad Z \xrightarrow{D} N(0, \Sigma)$$

$$AZ \xrightarrow{D} N(0, A \Sigma A')$$

$$\hat{\beta} = (x'x)^{-1} x'y$$

$$= \beta + (x'x)^{-1} x'e$$

$$\hat{\beta} - \beta = (x'x)^{-1} x'e \frac{n}{n^{1/2} n^{1/2}}$$

$$\sqrt{n} (\hat{\beta} - \beta) = (x'x)^{-1} x'e \frac{n}{\sqrt{n}}$$

$$\sqrt{n} (\hat{\beta} - \beta) = \left(\frac{x'x}{n} \right)^{-1} \left(\frac{x'e}{\sqrt{n}} \right)$$

$$plim \left(\sqrt{n} (\hat{\beta} - \beta) \right) = plim \left(\frac{x'x}{n} \right)^{-1} plim \left(\frac{x'e}{\sqrt{n}} \right)$$

$$E[x'x] = Q \quad \xrightarrow[p]{\quad} E[x'x]^{-1} \quad \xrightarrow[d]{\quad} N$$

Nota aparte.

$$plim \left(\frac{x'e}{\sqrt{n}} \right) \xrightarrow[N]{\quad} (0, \sigma_e^2 E[x'x])$$

$$E\left[\frac{X' \varepsilon}{\sqrt{n}}\right] = \frac{1}{\sqrt{n}} E[X' \varepsilon] = 0 \quad \text{por S2}$$

$$\text{Var}\left[\frac{X' \varepsilon}{\sqrt{n}}\right] = \frac{1}{n} X' \text{Var}(\varepsilon) X$$

$$= \frac{1}{n} \sigma_\varepsilon^2 X' I_n X$$

$$= \frac{1}{n} \sigma_\varepsilon^2 X' X$$

$$= \sigma_\varepsilon^2 \left(\frac{X' X}{n} \right)$$

$$= \sigma_\varepsilon^2 E[X' X]$$

$$\sqrt{n}(\hat{\beta} - \beta) = \text{plim} \left(\frac{X' X}{n} \right)^{-1} \text{plim} (Z)$$

$$\xrightarrow{P} Q^{-1} \xrightarrow{d} N(0, \sigma_\varepsilon^2 Q)$$

$$\xrightarrow{d} N(0, \sigma_\varepsilon^2 E(X' X)^{-1})$$

$$E[Q^{-1} Z] = 0 \quad \text{Var}[Q^{-1} Z] = Q^{-1} \underbrace{\text{Var}(Z)}_{\sigma_\varepsilon^2 Q} Q^{-1}$$

$$\begin{aligned} \text{Var}(Q^{-1} Z) &= \sigma_\varepsilon^2 \cancel{Q^{-1} Q} Q^{-1} = \sigma_\varepsilon^2 Q^{-1} \\ &= \sigma^2 E[X' X]^{-1} \end{aligned}$$

$$\hat{\beta} - \beta = (X'X)^{-1} X'E \frac{n}{n} = \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'E}{n} \right)$$

$\xrightarrow{P} \quad \xrightarrow{P}$
 $E[X'X]^{-1} \quad E[X'E]$
 $\neq 0$

$$\sqrt{n}(\hat{\beta} - \beta) = (X'X)^{-1} X'E \frac{n}{\sqrt{n} \cdot \sqrt{n}}$$

$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &= (X'X)^{-1} X'E \frac{n}{\sqrt{n}} \\ &= \left(\frac{X'X}{n} \right)^{-1} \underbrace{\left(\frac{X'E}{\sqrt{n}} \right)}_Z \end{aligned}$$

$$\begin{aligned} \text{plim}(\sqrt{n}(\hat{\beta} - \beta)) &= \text{plim} \left(\frac{X'X}{n} \right)^{-1} \text{plim}(Z) \\ &\xrightarrow{P \quad Q^{-1}} \quad \xrightarrow{N(0, \sigma_e^2 Q)} \\ &\quad \xrightarrow{\quad} N(0, \bullet) \end{aligned}$$

$$E[Q^{-1}Z] =$$

$$\text{Var}[Q^{-1}Z] = Q^{-1} \underbrace{\text{Var}(Z)}_{\sigma_e^2 Q} Q^{-1} = \sigma_e^2 E[X'X]^{-1}$$

~~$$G6. \epsilon \sim N(0, \sigma_\epsilon^2 I_n)$$~~



$$\hat{\beta} \sim N(\beta, \sigma_\epsilon^2 (X'X)^{-1}) \begin{matrix} \nearrow \mathcal{D} \\ \searrow F \end{matrix}$$

\

$$\epsilon \sim iid(0, \sigma_\epsilon^2 I_n)$$

$$n \rightarrow \infty$$

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma_\epsilon^2 E[X'X]^{-1})$$

$$\hat{\beta} \xrightarrow{d} N(\beta, \sigma_\epsilon^2 E[X'X]^{-1})$$

$$\rightarrow N(0, E[X'X]^{-1} \sigma_e^2 E[X'X] E[X'X]^{-1})$$

$$\sqrt{n} (\hat{\beta} - \beta) \xrightarrow{D} N(0, \sigma_e^2 E[X'X]^{-1})$$

$$S6b \quad \varepsilon \sim iid(0, \sigma_e^2) \\ + \quad n \rightarrow \infty$$

S1 lineal parametros

S2 $E[X'\varepsilon] = 0$ $Cov(X, \varepsilon) = 0$ ☺

S3 (colinealidad imperfecta $Rango(X) = K$)

S4 $Var(\varepsilon) = \sigma_e^2 I_n$ $Cov(\varepsilon_i, \varepsilon_j) = 0$

S5 X es fijo.

S6b. $\varepsilon \sim iid(0, \sigma_e^2) + n \rightarrow \infty$

Multicolinealidad (Supuesto S3)

$$y = x_1 \beta_1 + x_s \beta_s + \varepsilon$$

$$\hat{\beta}_1 = ((M_S X_1)' (M_S X_1))^{-1} (M_S X_1)' M_S y$$

$$\hat{\beta}_1 = [(M_S X_1)' (M_S X_1)]^{-1} (M_S X_1)' M_S [X_1 \beta_1 + X_S \beta_S + \varepsilon]$$

$$= \beta_1 + [(M_S X_1)' (M_S X_1)]^{-1} (M_S X_1)' M_S \varepsilon$$

$$E[\hat{\beta}_1 | X] = \beta_1 + \mathbf{0} \quad \rightarrow \text{part A2}$$

$$plim [\hat{\beta}_1] = \beta_1 + \underbrace{[(M_S X_1)' (M_S X_1)]^{-1} (M_S X_1)' M_S \varepsilon}_{A} \frac{n}{n}$$

Quê pasa si $\varepsilon \sim N(0, \sigma_\varepsilon^2)$? $= \sigma_\varepsilon^2 \frac{X'X}{n}$

Depende $n \rightarrow$ pequeno PILAS
 $\rightarrow \infty$!!

$$\hat{\beta} = \beta + (X'X)^{-1} X' \varepsilon$$

$$\hat{\beta} - \beta = (X'X)^{-1} X' \varepsilon \quad \frac{n}{\sqrt{n} \sqrt{n}}$$

$$\sqrt{n} (\hat{\beta} - \beta) = \underbrace{\left(\frac{X'X}{n} \right)^{-1}}_{\xrightarrow{p} E[X'X]^{-1}} \underbrace{\left(\frac{X' \varepsilon}{\sqrt{n}} \right)}_{\xrightarrow{d} N(0, \frac{\sigma_\varepsilon^2 (X'X)}{n})}$$

TCL $z_n = \frac{\varepsilon}{\sqrt{n}} \sim iid(0, \sigma_\varepsilon^2) \xrightarrow{d} N(0, \sigma_\varepsilon^2)$

$$A_n = X \xrightarrow{p} X$$

$$\left(\frac{X' \varepsilon}{\sqrt{n}} \right) = A' z \longrightarrow N(0, \frac{1}{n} \sigma_\varepsilon^2 (X'X))$$

$$\begin{aligned}\text{Var} \left(\frac{X' \varepsilon}{\sqrt{n}} \right) &= \frac{1}{n} X' \text{Var}(\varepsilon) X \\ &= \frac{1}{n} \sigma_\varepsilon^2 X' X\end{aligned}$$

$$z_n = \left(\frac{X' \varepsilon}{\sqrt{n}} \right) \xrightarrow{d} N(0, \underbrace{\sigma_\varepsilon^2 E[X'X]}_Q)$$

$$A = \left(\frac{X'X}{n} \right)^{-1} \xrightarrow{P_r} \underbrace{E[X'X]}_Q^{-1}$$

$$A z_n \xrightarrow{d} N(0, \quad)$$

$$E[A z_n] = Q E[z_n] = 0$$

$$\begin{aligned}\text{Var}[A z_n] &= A \text{Var}(z_n) A' = Q^{-1} \sigma_\varepsilon^2 \underbrace{Q Q^{-1}}_I \\ &= \sigma_\varepsilon^2 E[X'X]^{-1}\end{aligned}$$

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma_\varepsilon^2 E[X'X]^{-1})$$

$$(\hat{\beta} - \beta) \sim N(0, \sigma_e^2 (X'X)^{-1})$$

~~$$\epsilon \sim N(0, \sigma_e^2)$$~~

$$\epsilon \sim \text{iid}(0, \sigma_e^2) \text{ PERD}$$

$$n \rightarrow \infty$$

A3. Colinealidad Rango(X) = K

$$y = \overset{\text{DAE}}{X_1 \beta_1} + \overset{\text{AE}^2}{X_s \beta_s} + \epsilon$$

$$\underline{M_{X_s} y} = \underline{M_{X_s} X_1 \beta_1} + \cancel{\underline{M_{X_s} X_s \beta_s}} + M_{X_s} \epsilon$$

$$\underline{M_{X_s} y} = M_{X_s} X_1 \beta_1 + M_{X_s} \epsilon$$

$$\hat{\beta}_1 = [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} (M_{X_1} X_1)' \underline{M_{X_1} y}$$

$$\hat{\beta}_1 = [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} [M_{X_1} X_1]' [M_{X_1} X_1 \beta_1 + M_{X_1} \varepsilon]$$

$$= \underbrace{[(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} [M_{X_1} X_1]' [M_{X_1} X_1]}_{I_n} \underline{\beta_1}$$

$$+ [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} (M_{X_1} X_1)' M_{X_1} \varepsilon$$

$$= \beta_1 + [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} (M_{X_1} X_1)' M_{X_1} \varepsilon$$

$$E[\hat{\beta}_1] = \beta_1 \quad \text{INSESGADO}$$

$$\text{plim}[\hat{\beta}_1] = \beta_1 \quad \text{CONSISTENTE}$$

$$\text{Var}(\hat{\beta}_1) = \text{Var} \left(\underbrace{\beta_1 + [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} (M_{X_1} X_1)' M_{X_1} \varepsilon}_A \right)$$

$$= A \underbrace{\text{Var}(\varepsilon)}_{\varepsilon} A' = \sigma_\varepsilon^2 A A'$$

$$= \sigma_\varepsilon^2 [(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1} (M_{X_1} X_1)' M_{X_1}' (M_{X_1} X_1)$$

$$[(M_{X_1} X_1)' (M_{X_1} X_1)]^{-1}$$

$$\text{Var}(\hat{\beta}_1) = \sigma_e^2 \left[\underbrace{(M_{x_s} x_1)'}_{\underline{a}} \underbrace{(M_{x_s} x_1)}_{\underline{a}} \right]^{-1}$$

$$a. M_{x_s} x_1 = [I_n - P_{x_s}] x_1$$

$$= [I_n - x_s (x_s' x_s)^{-1} x_s'] x_1$$

$$= x_1 - x_s \underbrace{(x_s' x_s)^{-1} x_s' x_1}_{\alpha_{x_1, x_s}}$$

$$= x_1 - x_s \hat{\alpha}$$

$$= x_1 - \hat{x}_1$$

$$= \hat{u}_{x_1, x_s}$$

$$\rightarrow \boxed{x_1 = x_s \alpha + u}$$

$$\text{Var}(\hat{\beta}_1) = \sigma_e^2 \left(\underbrace{\hat{u}_{x_1, x_s}' \hat{u}_{x_1, x_s}}_{\text{SRC}_{x_1, x_s}} \right)^{-1}$$

$$= \sigma_e^2 \left(\text{SRC}_{x_1, x_s} \cdot \frac{\overbrace{x_1' x_1}^{\text{STC}_{x_1, x_s}}}{\underbrace{x_1' x_1}} \right)^{-1}$$

$$\begin{aligned}
 &= \sigma_e^2 (X_1' X_1)^{-1} \left(\frac{\overbrace{SRC_{x_1, x_s}}^{SRC_{x_1, x_s}}}{X_1' X_1} \right)^{-1} \\
 &= \sigma_e^2 (X_1' X_1)^{-1} \left(\frac{SRC_{x_1, x_s}}{\underbrace{STC_{x_1}}_{(1-R^2_{x_1, x_s})^{-1}}} \right)^{-1} \\
 &= \boxed{\frac{\sigma_e^2 (X_1' X_1)^{-1}}{1 - R^2_{x_1, x_s}}}
 \end{aligned}$$

$$R^2 = \frac{SEC}{STC} = 1 - \frac{SRC}{STC}$$

$$\boxed{\frac{SRC}{STC} = 1 - R^2_{x_1, x_s}}$$

$$\rightarrow \text{Var}(\hat{\beta}_1) = \sigma_e^2 \frac{(X_1' X_1)^{-1}}{1 - \underbrace{R^2_{x_1, x_s}}_{\nearrow \infty}}$$

$$= \frac{SRC}{n - K} \frac{(X_1' X_1)^{-1}}{1 - \underbrace{R^2_{x_1, x_s}}_{0-}} = \infty$$

$$y = X_1 \beta_1 + \varepsilon$$

$$y = X_1 \beta_1 + \cancel{X_2 \beta_2} + X_3 \beta_3 + \mathcal{U}$$

$$\text{Var}(\hat{\beta}_1) = \sigma_\varepsilon^2 (X_1' X_1)^{-1}$$

$$= \frac{SRC_1}{n-k_1} (X_1' X_1)^{-1}$$

$$\text{Var}(\beta_1^*) = \sigma_\varepsilon^2 \frac{(X_1' X_1)^{-1}}{1-R_{x_1, x_3}^2}$$

$$\geq \frac{SRC_2}{n-k_2} \frac{(X_1' X_1)^{-1}}{1-R_{x_1, x_3}^2}$$